The 10th International Competition of the Military Technical College

Artificial Intelligence Competition

(AIC-4)

AIC
ARTIFICIAL
INTELLIGENCE
COMPETITION

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1 Introduction

AIC-4 is one of the best Egypt National Artificial Intelligence Competitions brought to you by the Military Technical College and the Applied Innovation Center (AIC) of the Ministry of Communications and Information Technology.

Aerial object tracking is a pivotal computer vision task, critical for applications from autonomous UAV navigation to defense and environmental monitoring. However, real-world deployment demands trackers that are not only accurate but computationally efficient as well. This competition challenges the global AI community to pioneer the next generation of robust, lightweight tracking solutions that can perform in dynamic, continuous, and stochastic aerial environments. Join us in advancing the frontier of practical, high-performance aerial vision.

Any issues not covered by these published rule sets will be addressed on a case-by-case basis by the competition board. If you have any questions and comments, kindly direct them to competition@mtc.edu.eg. English is the official language of the competition, and each team must submit their deliverables in English.

2 Competition Goals

The goal of this competition is to design a **single-object aerial tracker** that achieves the highest tracking accuracy while maintaining minimal computational complexity. Specifically, participants are encouraged to:

- Maximize tracking precision and success rate on challenging aerial sequences.
- Handle fast camera motion, scale variation, occlusion, and small object size.
- Minimize tracking model size, number of parameters, and inference Flops.
- Develop trackers suitable for real-time deployment on UAV platforms.

3 Datasets

The competition will leverage several publicly available **single-object aerial tracking (SOT)** datasets. All datasets provide frame-by-frame bounding box annotations and follow the standard SOT protocol, where the tracker is initialized using the ground-truth target location in the first frame.

Table 1: Summary of Single Object Aerial Tracking Datasets

Dataset	#Sequences	#Frames	Challenges
UAV123	123	∼110K	Scale, Motion, Viewpoint
UAV20L	20	∼58K	Occlusion, Out-of-view
DTB70	70	∼50K	Fast motion, Blur
UAVTrack112	112	∼30K	Small targets, Camera motion

3.1 UAV123

- URL: https://ivul.kaust.edu.sa/benchmark-and-simulator-uav-tracking-dataset
- Description: UAV123 is a widely used aerial single-object tracking benchmark consisting of 123 video sequences and approximately 110,000 annotated frames. The dataset covers diverse UAV tracking challenges, including **but not limited to** fast camera motion, scale variation, occlusion, and viewpoint changes.

3.2 UAV20L

- URL: https://ivul.kaust.edu.sa/benchmark-and-simulator-uav-tracking-dataset
- Description: UAV20L is a long-term single-object tracking subset derived from UAV123. It contains 20 long video sequences designed to evaluate tracker robustness under prolonged target absence, heavy occlusion, and out-of-view scenarios.

3.3 DTB70

- URL: https://github.com/flyers/drone-tracking
- Description: The Drone Tracking Benchmark (DTB70) consists of 70 aerial video sequences captured by UAVs. It focuses on challenging conditions such as motion blur, abrupt camera movement, low resolution targets, and background clutter, making it suitable for evaluating robust SOT algorithms.

3.4 UAVTrack112

- URL: https://github.com/vision4robotics/SiamAPN
- Description: UAVTrack112 is a recent single-object tracking dataset specifically designed for UAV scenarios. It contains 112 video sequences with small object sizes, complex backgrounds, and significant camera motion, emphasizing realistic aerial tracking challenges.

4 Evaluation Metrics

4.1 Tracking Accuracy Metrics

- **Precision:** Distance between predicted and ground-truth target centers.
- **Success Rate (AUC):** Intersection-over-Union (IoU) based overlap evaluation.
- **Normalized Precision:** Scale-invariant center error.
- **Robustness:** Number of tracking failures (for long-term sequences).

Standard evaluation scripts will be provided.

4.2 Efficiency Metrics

- **Number of Parameters**
- **Inference FLOPs**
- **Runtime Latency** (measured on standardized hardware)
- **Memory Footprint** (model size)

4.3 Dataset Licensing and Usage Policy

All public datasets used in this competition are released for academic and research purposes. Participants are permitted to use these datasets for model training and development in accordance with their original licenses.

Participants are **not allowed** to redistribute any dataset content as part of their submissions. Only trained models, source code, and documentation may be submitted.

Final evaluation will be conducted on a private dataset owned by the organizers. No data from the private evaluation set will be released to participants.

5 Submission

5.1 Important notes

- The tracker is initialized using the ground-truth bounding box in the first frame.
- No re-initialization is allowed during evaluation unless explicitly stated.
- Online-only tracking is required (no future frame access).

5.2 Submission Guidelines

- Trained model files and code.
- Instructions for inference and evaluation.
- A short technical report (max 4 pages).

5.3 Final Scoring Function

The final competition score balances tracking accuracy and model efficiency:

$$S_{\text{acc}} = w_1 \cdot \text{AUC} + w_2 \cdot \text{Precision}$$

$$S_{\text{eff}} = w_3 \cdot \tilde{\text{FLOPs}} + w_4 \cdot \tilde{\text{Params}} + w_5 \cdot \tilde{\text{Latency}} + w_6 \cdot \tilde{\text{ModelSize}}$$

$$\text{Final Score} = S_{\text{acc}} - \lambda \cdot S_{\text{eff}},$$

where all efficiency metrics are normalized to $[0, 1]$ across submissions. The normalization will be performed using predefined reference budgets as $\tilde{m} = \min\left(1, \frac{m}{m_{\max}}\right)$, where $m_{\max}$ is a fixed budget per metric announced before the competition. Example is shown in Table. 2 (**It will be changed later before the actual competition**).

Table 2: Predefined Efficiency Budgets for Model Evaluation

Metric	**Maximum Allowed Value**
Computational Complexity (FLOPs)	20 GFLOPs
Number of Parameters	10 Million
Inference Latency	30 ms
Model Size (Disk Footprint)	40 MB

6 Awards

Talented Artificial Intelligence enthusiasts compete for the fame and the prize pool of 300,000 Egyptian pounds. Prizes will be awarded to the top-performing teams or individuals as the following:

- 1st Place: 150,000 Egyptian pounds.
- 2nd Place: 100,000 Egyptian pounds.
- 3rd Place: 50,000 Egyptian pounds.

7 Competition Rules

7.1 Eligibility

Undergraduate students registered in any of the Egyptian or international universities.

7.2 Team members

Participants are encouraged to form teams of 3-5 members.

7.3 Registration

- Participants must register for the competition through https://aic.conferences.ekb.eg/
- Registration requires:
  - Team Name / Members / Proof of Identity
  - University Name / Faculty Name
  - Proof of enrollment in the university.
  - Official agreement letters for the participant team from their university
- Late registrations may not be accepted.

7.4 Prohibited activities

- Each participant can only join one team. Multiple accounts per user are not permitted, and neither is collaboration or membership across multiple teams. Individuals and their submissions originating from multiple accounts will be immediately disqualified.
- Sharing of any related materials of the competition outside each individual team.
- Participants must not use any illegal or unethical method to complete challenges.
- The leaderboard will be announced in anonymous names till the organizing team verifies the final release of results.
- Violating these rules may result in disqualification.

7.5 Disclaimers

- The competition organizers are not responsible for any loss or damage to participants' equipment or data.
- The competition organizers reserve the right to modify the rules and format of the competition at any time.
- By participating in the competition, participants agree to abide by these rules and any decisions made by the competition organizers.

8 Competition Deadlines

Table 3: Competition Submission Deadlines

Activity	**Date**
AIC 2026 Rules provided to entrants	10th February 2026
Datasets available for participating teams	10th February 2026
Phase One Starting	10th February 2026
Registration Closed	22nd February 2026
Phase One Submission Deadline	11th April 2026
Qualified Teams for Phase Two (20 Teams)	18th April 2026
Registration Documents Submission (IDs)	18th April 2026
Phase Two Starting	19th April 2026
Phase Two Submission Deadline	11th May 2026
Qualified Teams for Final Round (5 Teams)	18th May 2026
Acceptance from the Technical Committee	4th June 2026
Final Round (Field Demonstration)	22th June to 3rd July 2026
Final Round	11th to 16th July 2026